

Keywords: ionic bonding, electrostatic, ions

**DO NOW:** Let's do a quiz about your previous lesson. Quiz 1 □ 5



# LOb: Understand how ionic bonding forms.

**Keywords:** ionic bonding,  
electrostatic, ions

## Learning Outcomes:

- Describe an ionic bond
- Display and describe a reaction producing an ionic bond

Friendship is the strongest  
bond in the world.

Jeremy Taylor

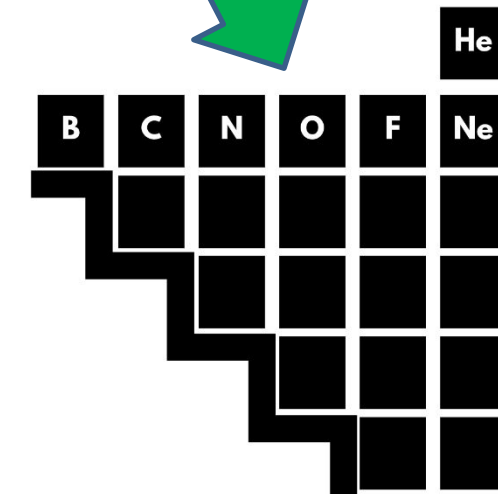
**Did you know...** This quote is wrong. Scientifically the strongest bonds are called ionic and covalent bonds. They happen when certain elements chemically join together.



This is an ionic compound

CuS

4

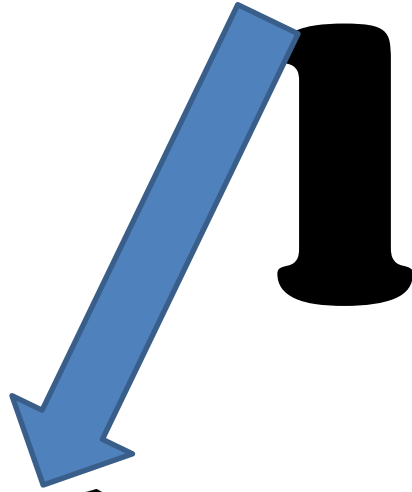
A partial periodic table diagram showing the first four rows of elements. The elements are represented by black boxes with white text. The first row contains He. The second row contains B, C, N, O, F, Ne. The third row contains four empty boxes. The fourth row contains three empty boxes. The fifth row contains two empty boxes. The sixth row contains one empty box.

|   |   |   |   |   |    |
|---|---|---|---|---|----|
|   |   |   |   |   | He |
| B | C | N | O | F | Ne |
|   |   |   |   |   |    |
|   |   |   |   |   |    |
|   |   |   |   |   |    |
|   |   |   |   |   |    |



## Is this an ionic compound?

# HC



|   |   |   |   |   |    |
|---|---|---|---|---|----|
| B | C | N | O | F | He |
|   |   |   |   |   |    |
|   |   |   |   |   |    |
|   |   |   |   |   |    |
|   |   |   |   |   |    |
|   |   |   |   |   |    |

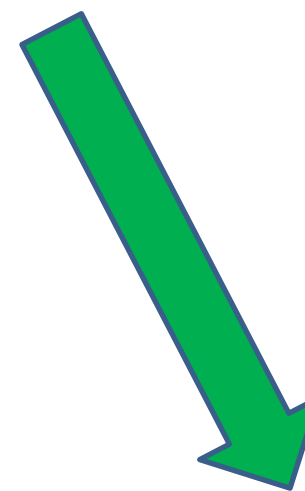
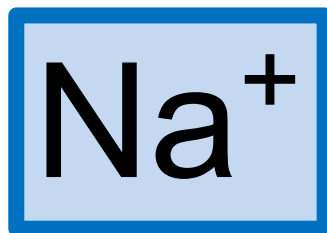
# What are the charges in this compound?

1. Metal = positive, group number tells you the size
2. Non-metal = negative, 8 – group number tells you the size

# NaCl



Metal, group 1



Non-metal, 8- group 7 = 1



*Note. Compound have an overall charge of zero so charges must cancel each other out.*

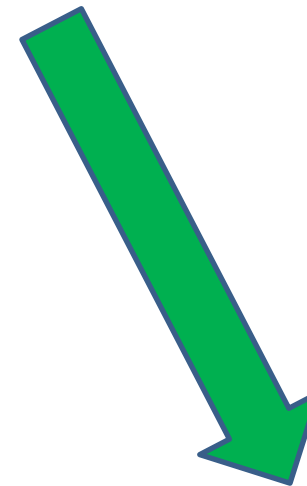
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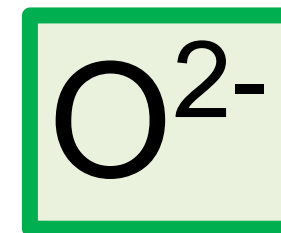
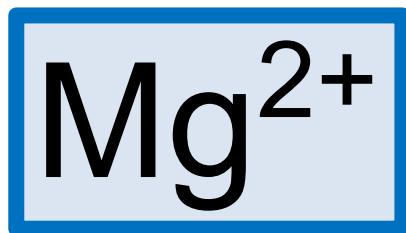
**Mg**

**O**



Metal, group 2

Non-metal, 8- group 6 = 2

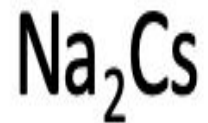


*Note. Compound have an overall charge of zero so charges must cancel each other out.*

**YOU DO: Complete the questions on your sheet**



hydrochloric acid



lithium chloride

magnesium oxide

silver nitrate

water

hydrogen bromide



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## Learning Outcomes:

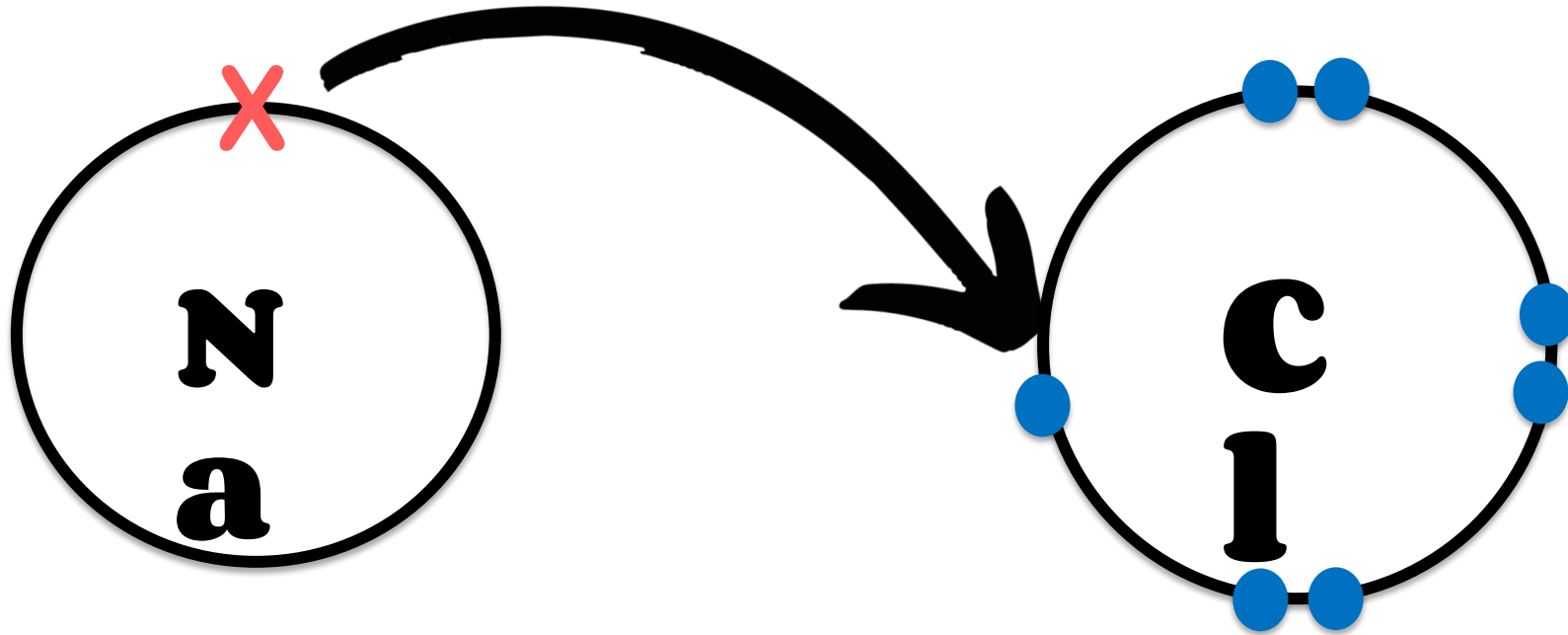
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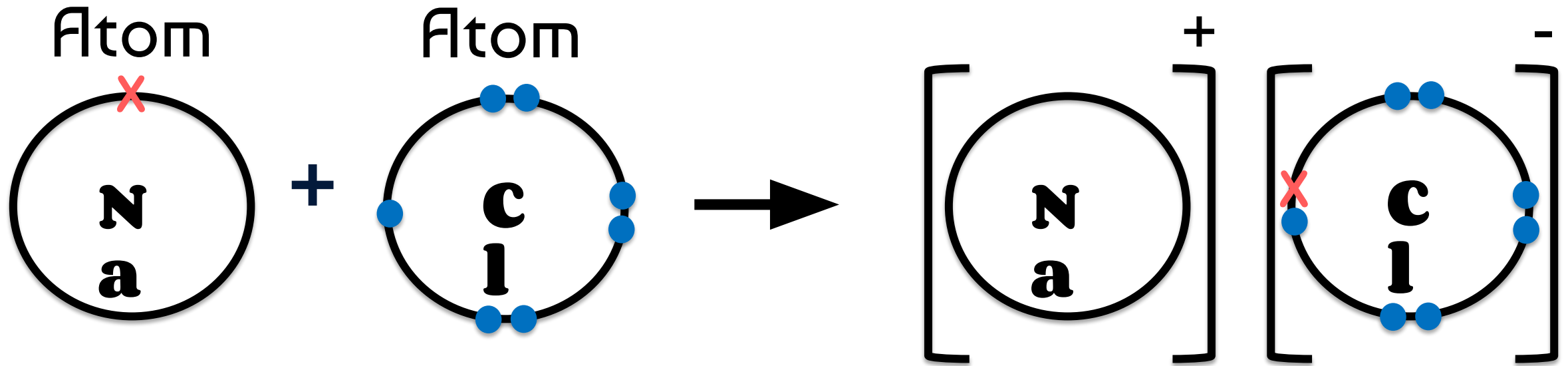
Jeremy Taylor

**Did you know...** This quote is wrong. Scientifically the strongest bonds are called ionic and covalent bonds. They happen when certain elements chemically join together.

Ionic bonding involves a **transfer of electrons** from the metal to the non-metal.

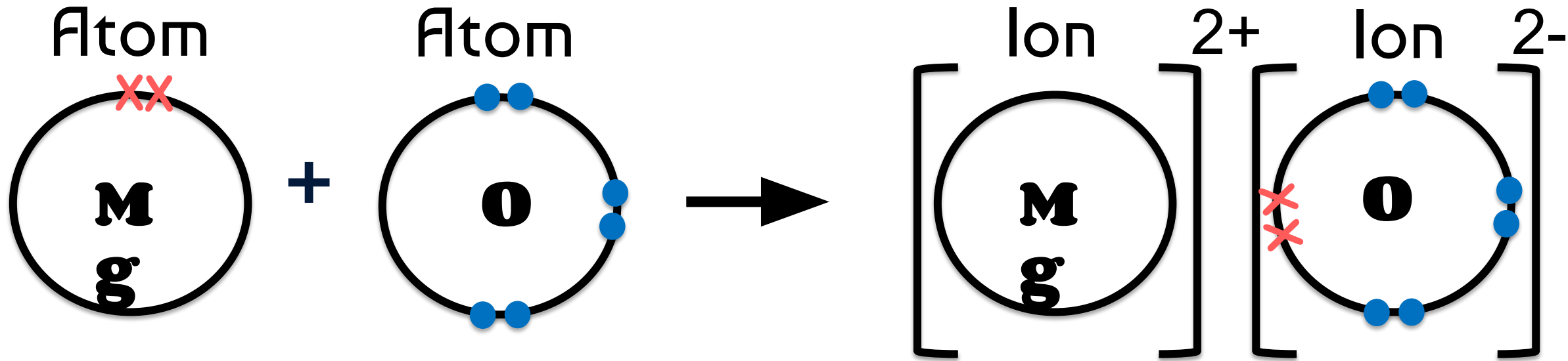


# Here is an ionic bond diagram



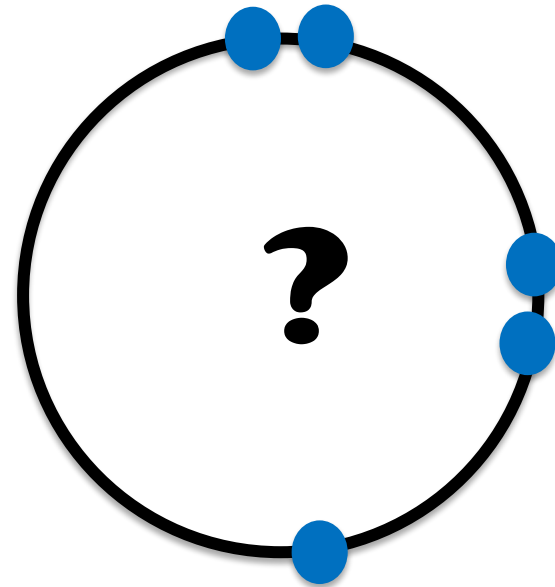
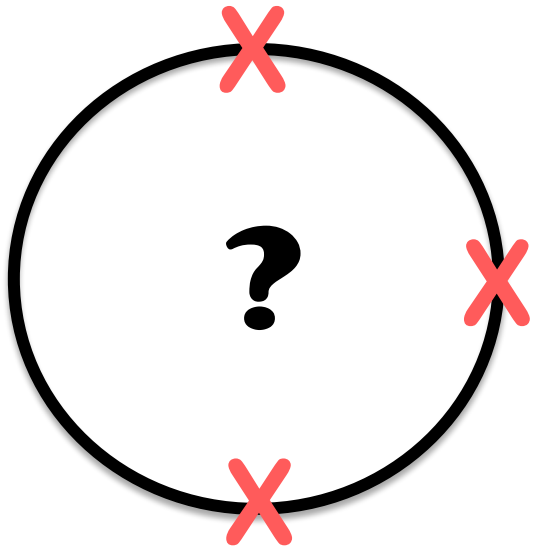
- ☐ Dot and cross
- ☐ Brackets
- ☐ Charges
- ☐ Outer shell ONLY

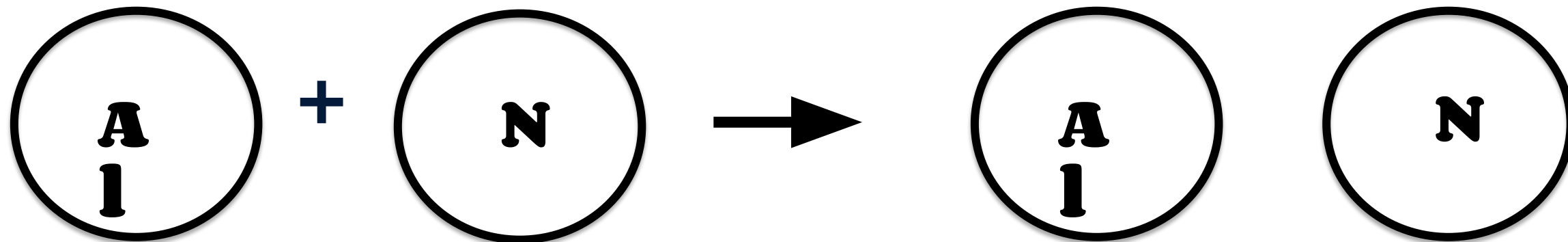
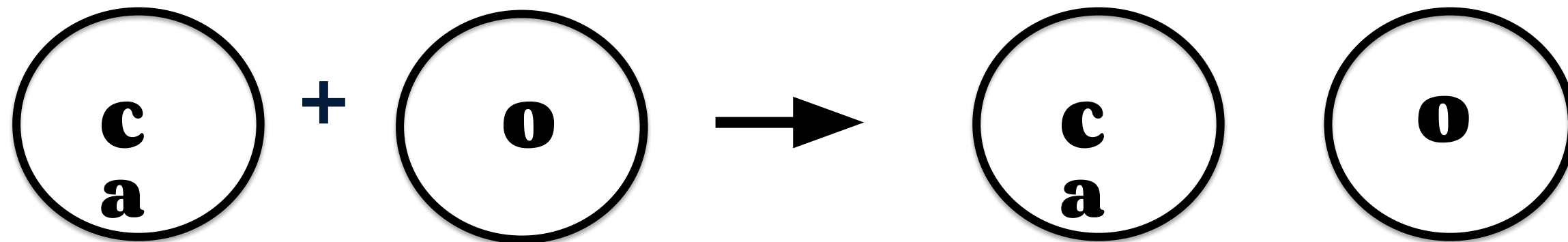
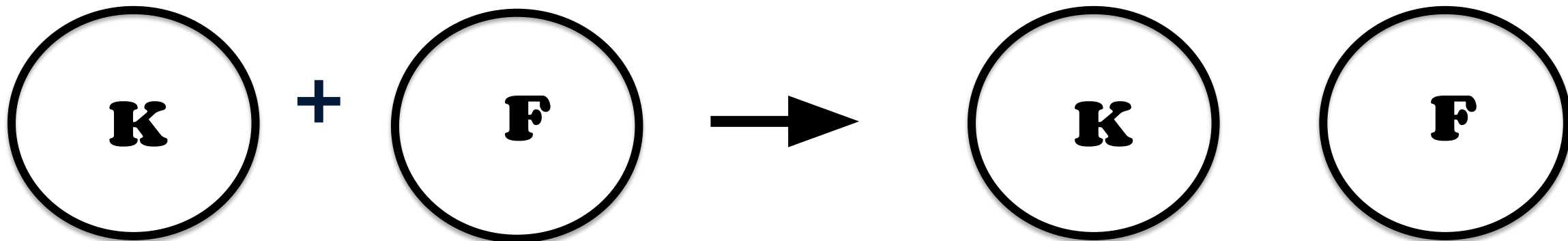
# Magnesium reacting with oxygen to form magnesium oxide



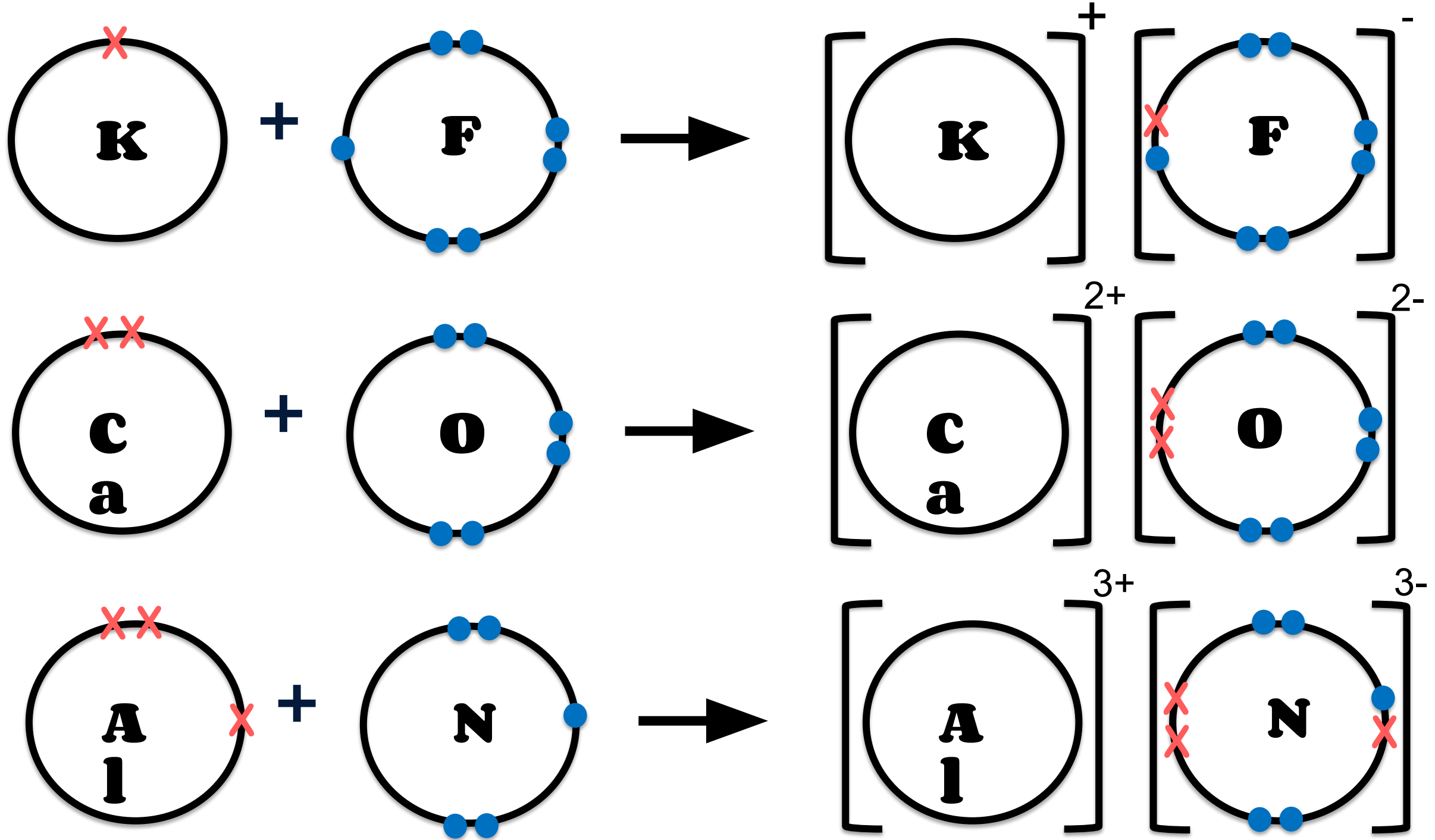
- ☐ Dot and cross
- ☐ Brackets
- ☐ Charges
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# YOU DO: Complete the ionic bond diagrams



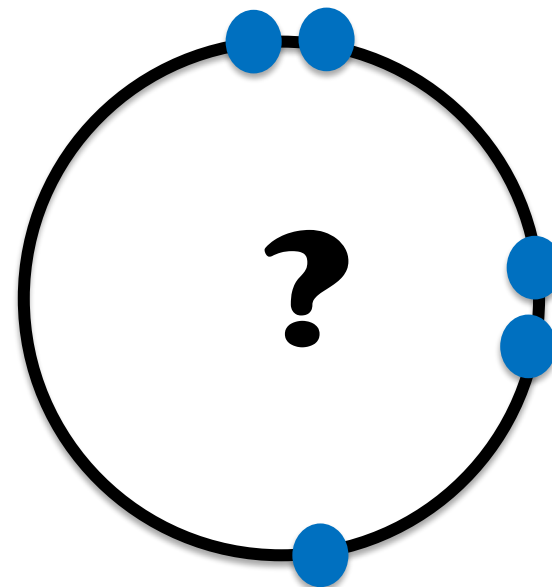
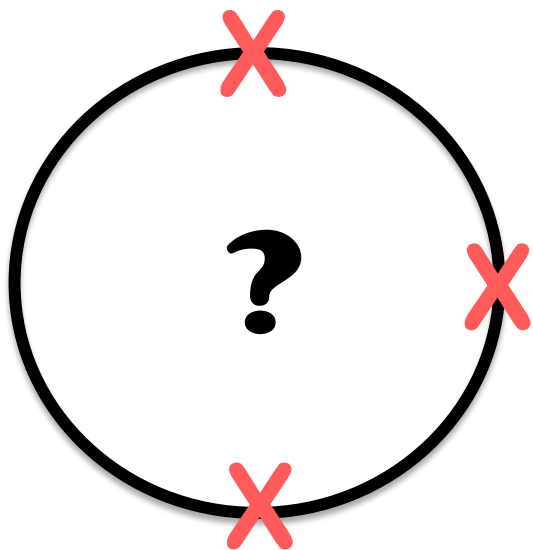




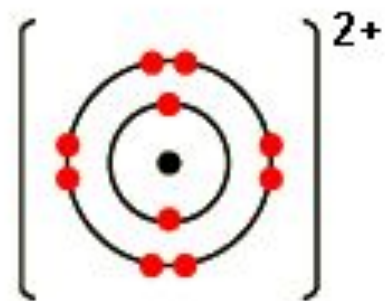


# YOU DO: Complete the ionic bond diagrams for these substances

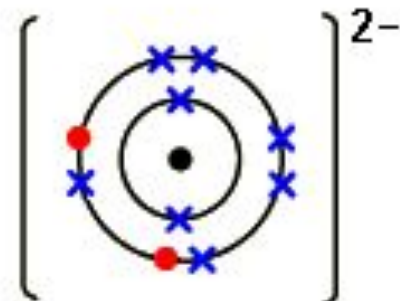
- a) magnesium oxide
- b) Calcium fluoride
- c) Sodium sulphide
- d) aluminium oxide
- e) calcium chloride



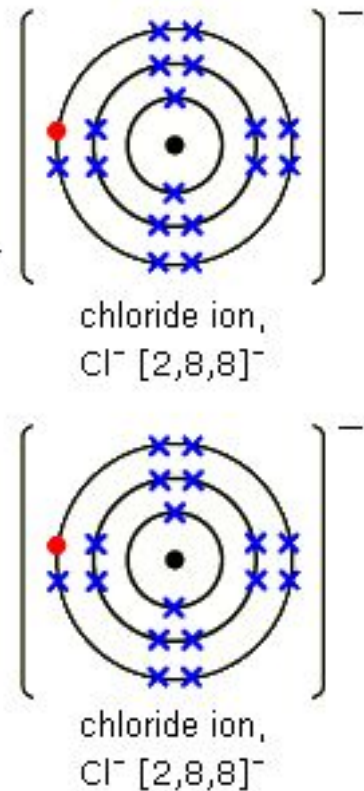
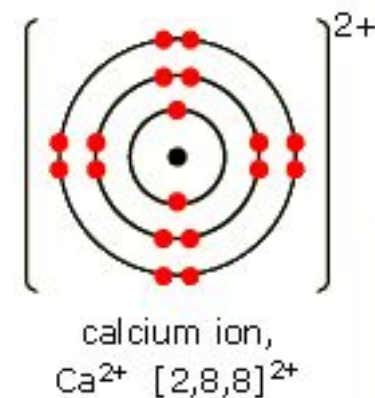
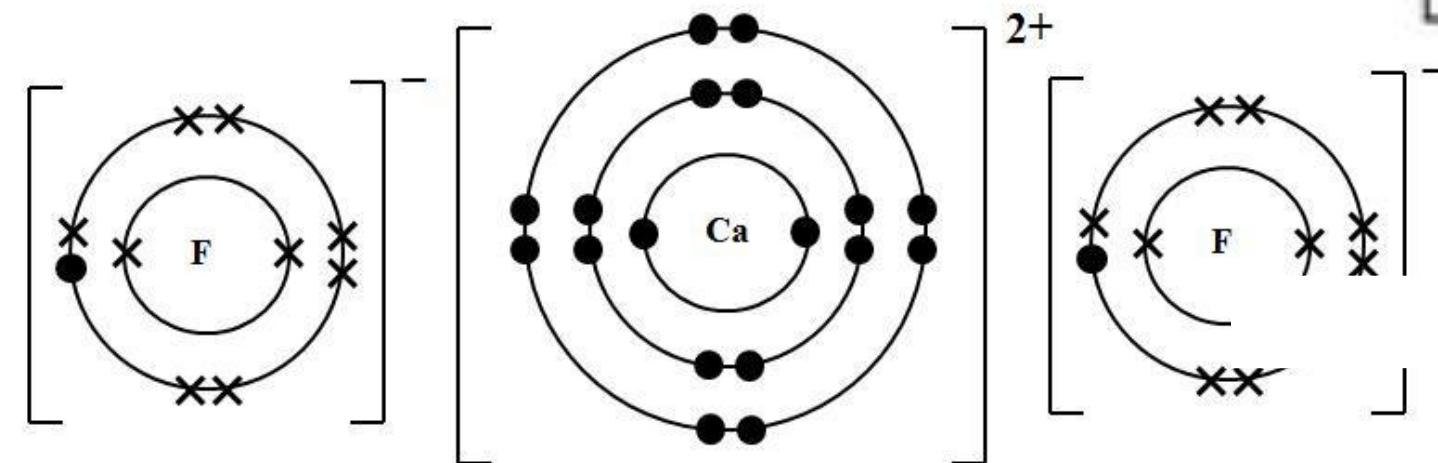
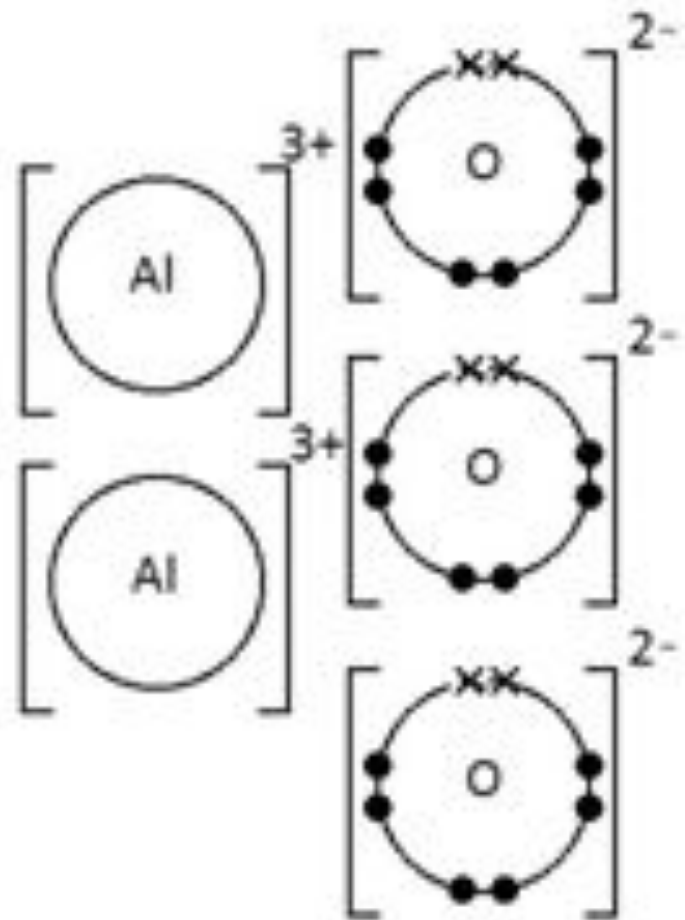
# Review



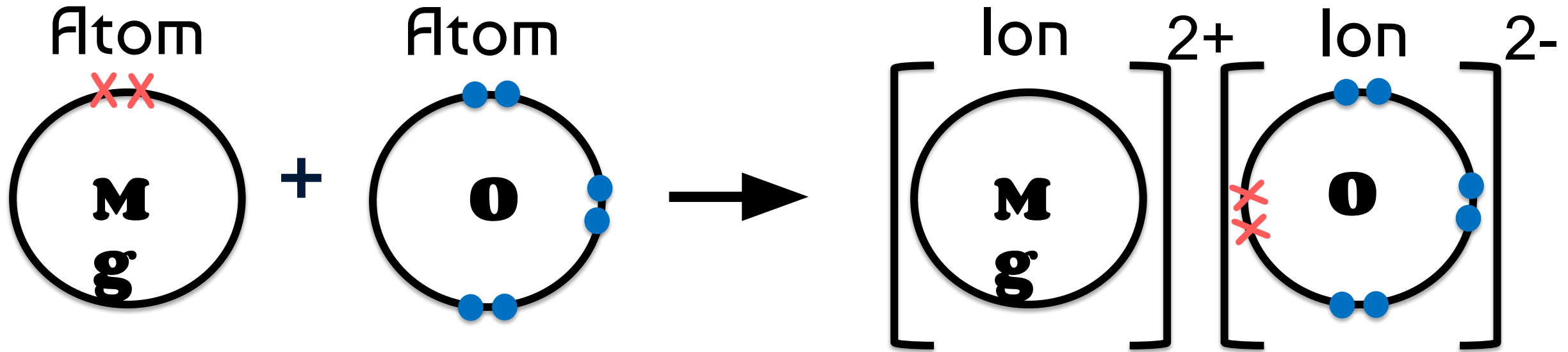
magnesium ion,  
 $\text{Mg}^{2+}$  [2,8] $^{2+}$



oxide ion,  
 $\text{O}^{2-}$  [2,8] $^{2-}$



# Describing what is made



Magnesium  
atom  
loses 2  
electrons

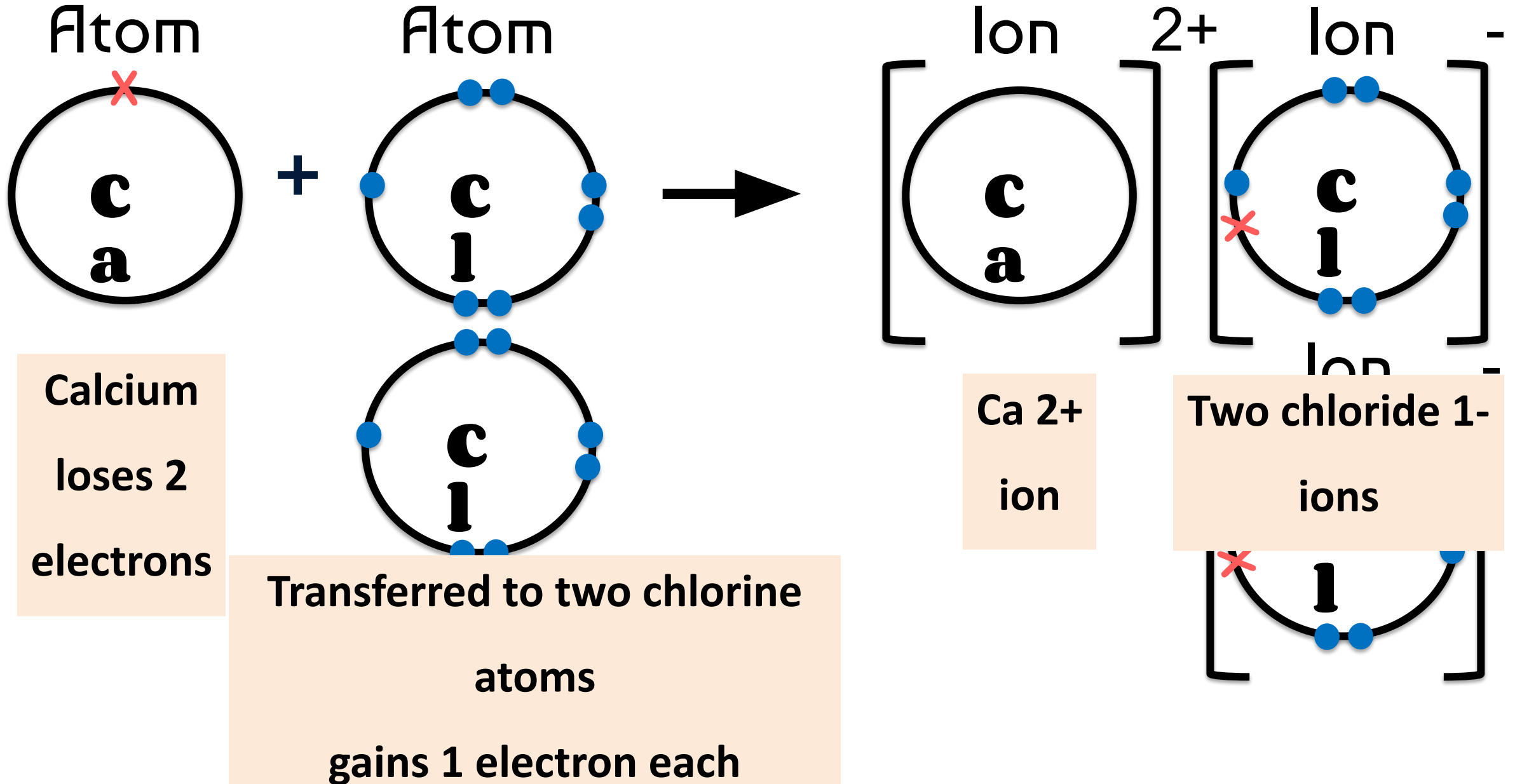
Transferred to oxygen  
atom,  
gains 2  
electrons

Mg 2+  
ion

Oxide 2-  
ion

Both have full outer shells

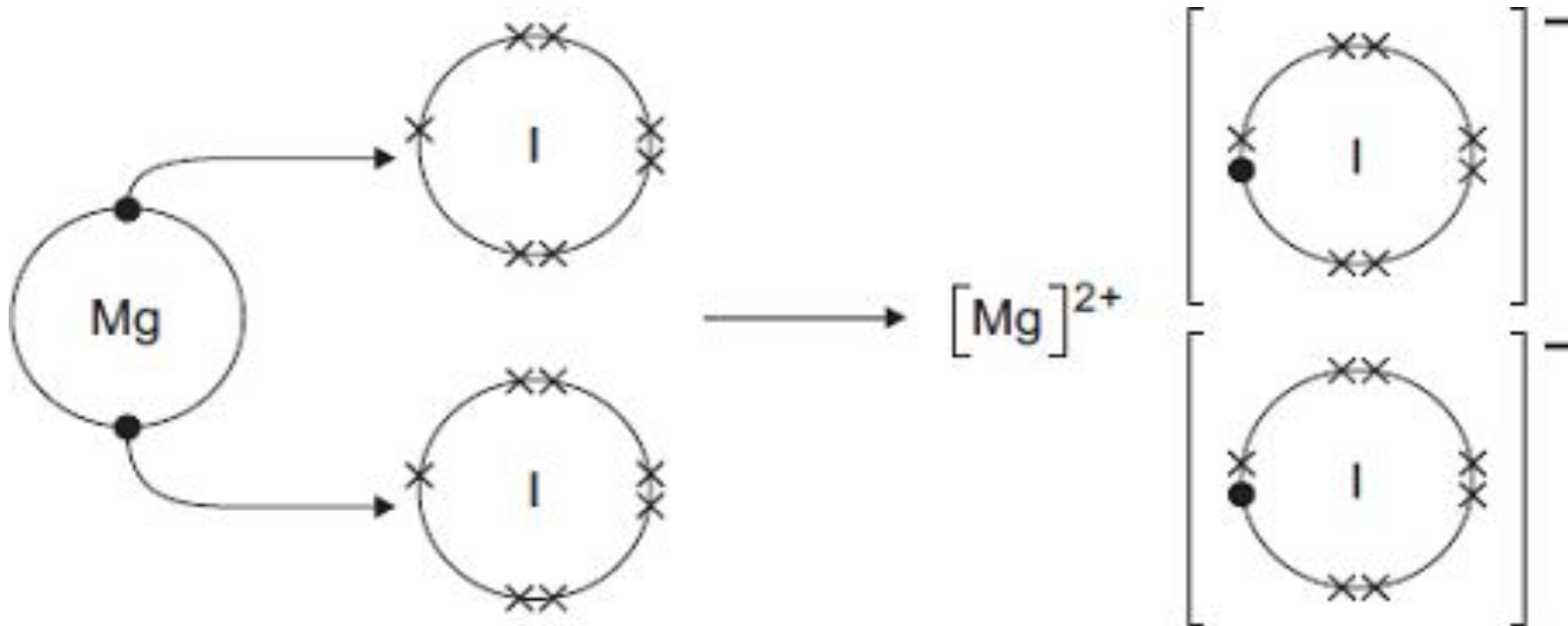
# Calcium reacting with chlorine to form calcium chloride



## Exam Question

Describe, as fully as you can, what happens when magnesium reacts with iodine to make magnesium iodide.

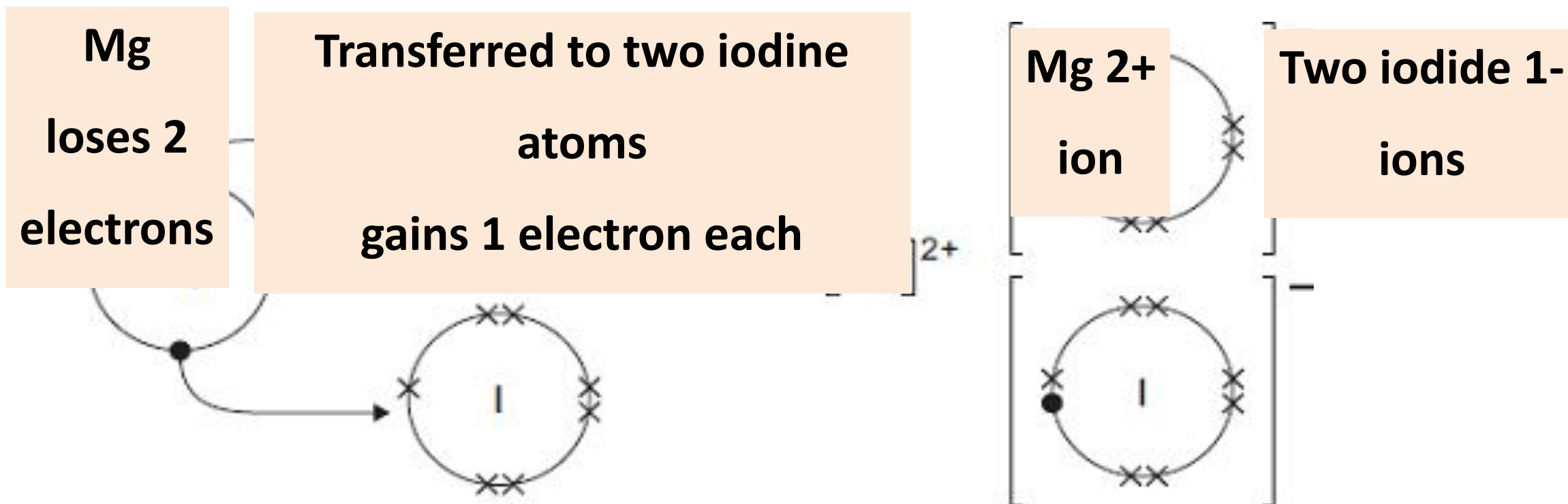
To gain full marks you should use the words atom, electron and ion in your answer.



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**YOU DO: Complete the questions on your sheet**



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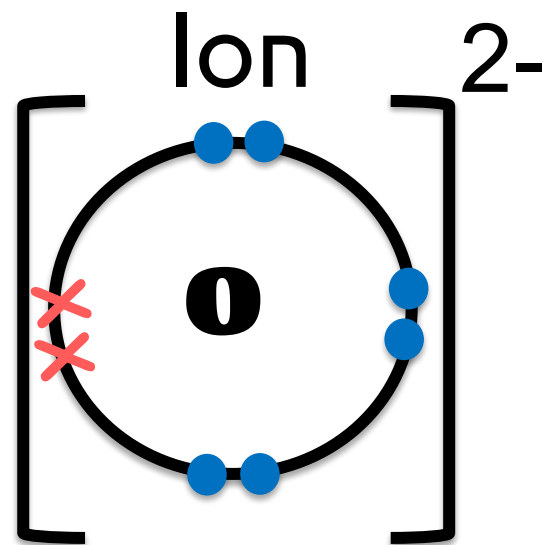
# keyfacts

 **Ionic bonding** is a **transfer of electrons** between a metal and non-metal

 Ionic bonding has an **electrostatic attraction** between the **positive** and **Negative ions**.

 Ionic bond drawings have **brackets**, **dot and cross charges** and **outer shell** only.

 **Describe** how many electrons transferred, and what is created.



# Plenary

1. An ion is an atom that has \_\_\_\_\_
2. If it gains electrons it will have a \_\_\_\_ charge
3. If it loses electrons it will have a \_\_\_\_ charge
3. An ion has a \_\_\_\_ outer shell of electrons
4. You are allowed 2 electrons on the first shell, \_\_\_\_ on the second and third shell and \_\_\_\_ on the fourth
5. If this an ion. Explain your answer.

